Exponential Generating Functions

Definitions, operations, and the multiplication principle

1. Ordinary and exponential generating functions

Given a sequence $a_0, a_1, a_2, \ldots$, its *ordinary generating function* (OGF) is

$$A_{\text{ord}}(x) = \sum_{n \geq 0} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots .$$

Its *exponential generating function* (EGF) is

$$A(x) = \sum_{n \geq 0} a_n \frac{x^n}{n!} = a_0 + a_1 x + a_2 \frac{x^2}{2!} + a_3 \frac{x^3}{3!} + \cdots .$$

We treat these as formal power series over a field of characteristic zero; questions of analytic convergence are not needed.

Why use an EGF? For the sequence $0!, 1!, 2!, 3!, \ldots$, the OGF is the formal series $\sum_{n \geq 0} n! x^n$, whereas the EGF has the simple expression

$$\sum_{n \geq 0} n! \frac{x^n}{n!} = \sum_{n \geq 0} x^n = \frac{1}{1-x}.$$

More generally, ordinary generating functions naturally encode convolutions $c_n = \sum_{k=0}^{n} a_k b_{n-k}$, while exponential generating functions naturally encode the binomially weighted convolutions

$$c_n = \sum_{k=0}^{n} \binom{n}{k} a_k b_{n-k}.$$

2. Common sequences

Sequence a_n	EGF	OGF
1	e^x	$\dfrac{1}{1-x}$
n	xe^x	$\dfrac{x}{(1-x)^2}$
n^2	$(x+x^2)e^x$	$\dfrac{x+x^2}{(1-x)^3}$
$n!$	$\dfrac{1}{1-x}$	$\displaystyle\sum_{n \geq 0} n! x^n$
$\binom{N}{n}$	$\displaystyle\sum_{j=0}^{N} \binom{N}{j} \frac{x^j}{j!}$	$(1+x)^N$
2^n	e^{2x}	$\dfrac{1}{1-2x}$

Here n starts at 0. In the binomial row, N is fixed and $\binom{N}{n} = 0$ for $n > N$.

3. Operations on exponential generating functions

Let $A(x) = \sum_{n\geq 0} a_n x^n/n!$.

Shift right. The EGF of $0, a_0, a_1, a_2, \ldots$ is the formal integral with zero constant term:

$$\int_0^x A(t)\,dt = \sum_{n\geq 0} a_n \frac{x^{n+1}}{(n+1)!}.$$

Shift left. The EGF of $a_1, a_2, a_3, \ldots$ is

$$A'(x) = \sum_{n\geq 0} a_{n+1} \frac{x^n}{n!}.$$

Add or subtract one from every term. The sequences $(a_n + 1)_{n\geq 0}$ and $(a_n - 1)_{n\geq 0}$ have EGFs $A(x) + e^x$ and $A(x) - e^x$, respectively.

Multiply every term by two. The sequence $(2a_n)_{n\geq 0}$ has EGF $2A(x)$. More generally, $(\lambda a_n)_{n\geq 0}$ has EGF $\lambda A(x)$ for a constant λ.

4. The multiplication principle

Suppose

$$A(x) = \sum_{n\geq 0} a_n \frac{x^n}{n!}, \qquad B(x) = \sum_{n\geq 0} b_n \frac{x^n}{n!}.$$

Then

$$\boxed{A(x)B(x) = \sum_{n\geq 0} \left(\sum_{k=0}^{n} \binom{n}{k} a_k b_{n-k} \right) \frac{x^n}{n!}.}$$

Indeed, collecting terms of total degree n gives

$$[x^n]\, A(x)B(x) = \sum_{k=0}^{n} \frac{a_k b_{n-k}}{k!(n-k)!} = \frac{1}{n!} \sum_{k=0}^{n} \binom{n}{k} a_k b_{n-k}.$$

Thus the coefficient of $x^n/n!$ in the product is $\sum_{k=0}^{n} \binom{n}{k} a_k b_{n-k}$.

Combinatorial interpretation. Suppose a_n counts structures called "widgets" on a given set of n labels, and b_n counts structures called "doodles" on such a set. Write $[n] = \{1, 2, \ldots, n\}$. The product $A(x)B(x)$ is the EGF for triples (S, u, v), where

- $S \subseteq [n]$;
- u is a widget on S;
- v is a doodle on $[n] \setminus S$.

If $|S| = k$, there are $\binom{n}{k}$ choices for S, then a_k choices for u and b_{n-k} choices for v. Summing over k proves the formula. The binomial coefficient records how the labels are split between the two structures; it is absent from the ordinary generating function product.

5. Words over a two-letter alphabet

All words in A and B

There are 2^n words of length n over $\{A, B\}$, so their EGF is

$$\sum_{n\geq 0} 2^n \frac{x^n}{n!} = e^{2x}.$$

The multiplication principle gives the same answer. There is exactly one word consisting only of A's of each length, and exactly one consisting only of B's. Both classes therefore have EGF e^x, and their labelled product has EGF

$$e^x e^x = e^{2x}.$$

The underlying bijection sends a word to the triple (S, u, v) in which $S \subseteq [n]$ is the set of positions occupied by A, $u = A^{|S|}$, and $v = B^{n-|S|}$. Conversely, the triple reconstructs the word by putting A in the positions of S and B in the remaining positions. Equivalently,

$$2^n = \sum_{k=0}^{n} \binom{n}{k}.$$

An even number of A's and an even number of B's

The EGF for an even-sized set of positions all occupied by A is

$$E(x) = \sum_{j\geq 0} \frac{x^{2j}}{(2j)!} = \frac{e^x + e^{-x}}{2}.$$

The same expression applies to B. Thus the EGF for words in which *both* letter counts are even is

$$\boxed{E(x)^2 = \left(\frac{e^x + e^{-x}}{2}\right)^2 = \frac{e^{2x} + 2 + e^{-2x}}{4}.}$$

If c_n denotes the number of such words of length n, this gives

$$c_n = \begin{cases} 1, & n = 0, \\ 0, & n \text{ is odd}, \\ 2^{n-1}, & n \geq 2 \text{ is even}. \end{cases}$$

For example, the valid words of length 2 are AA and BB.

Note on the transcription. The final handwritten formula, $(e^{2x} + e^{-2x})/2$, counts all words of even total length. It has been corrected above to impose even counts of both letters. In the multiplication principle, the doodle is on the complement of S and therefore has size $n - k$, rather than k. The zero constant term in the right-shift integral has also been made explicit.